



Camélia Ressam

MRI physics PhD candidate (NeuroSpin CEA Paris-Saclay,
Paris-Saclay University)

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Education

- **Physics and Medical Imaging PhD**, Paris-Saclay University & CEA Paris-Saclay 2023 – Today
« Ultra-High Field MRI for Human Brain Metabolism: CEST Imaging from 7 T to 11.7 T and Perspectives for Artificial Intelligence Integration », under direction of Dr. Luisa Ciobanu
Doctoral fellowship, **ranked first**
Attended lectures from the **MVA Master** (Paris-Saclay): Deep Learning in Practice, Deep Learning for Medical Imaging
- **Master 2, Computational Neurosciences and NeuroEngineering**, Paris-Saclay University 2022 - 2023
Double Degree, with highest honor
- **Master 2, Engineering**, CentraleSupélec, Paris-Saclay 2022 - 2023
Healthcare and Biomedical Services specialization
GPA: 4.2/4.3
- **Exchange Semester**, TU Wien, Austria 2022
Biomedical Engineering Master 2, Summer semester
- **L3 et Master 1, Engineering** – École Centrale de Marseille, Marseille 2019 - 2021
- **Classe Préparatoire aux Grandes Écoles** – Lycée Louis-le-Grand, Paris 2017 - 2021
PCSI – PC*

Professional Experiences

- **Neuroimaging research internship** – Neurospin, CEA Paris-Saclay 2023
« Assessing the robustness and reliability of fMRI-based biomarkers for a Small-Vessel Disease »
6-month internship under direction of Dr. Davide Boido
- **Deep Brain Stimulation internship** – Boston Scientific, France 2021
Design and teaching of written and oral training sessions for doctoral students and engineers on brain mapping applied to Parkinson's disease therapies.
5-month internship under direction of Gregg Malarde, Product Manager
- **Biomechanical research internship** – Institut des Sciences du Mouvement, Marseille 2021
« Study of the biomechanical and physiological adaptation of the locomotor pattern to variations in body weight », CNES project
6-month internship under direction of Dr. Pascale Chavet

Programming Languages and Languages

Programming languages: Python, MATLAB, R, C++, LaTeX

Languages: - French (Mother tongue) - English (C1, TOEIC 980/990)
- Spanish (B1) - Chinese (Beginner)

Publications and Conferences

Multi-Pool CEST MRI at 7 T reveals complementary metabolic and microstructural signatures to 18F-FDG PET across aging and in Alzheimer's disease, **Camélia Ressam** et al. (in preparation)

Multi-Contrast Human Brain CEST MRI at 11.7 T : First In Vivo Demonstration, **Camélia Ressam** et al., Magnetic Resonance in Medicine, 2026 (in review)

In Vivo Human Brain CEST MRI at 11.7T: First Results, **Camélia Ressam** et al., ISMRM 2026, Cape Town, South Africa. Oral communication. Magna Cum Laude Merit Award

CEST Perspectives for Human Brain Applications at 11.7 T, **Camélia Ressam** et al., ISMRM 2025, Honolulu, Hawaii, USA. Poster

Characterization of Metabolic Changes in Healthy Aging Using High-Field CEST MRI, Thaddée Delebarre, **Camélia Ressam** et al., EMIM 2024, Porto, Portugal. Poster

Science Outreach

- **Femmes en Sciences** 2025
Invited speaker at a science outreach conference organized by the National Museum of Natural History and the CEA. I shared my journey and experience as a doctoral student alongside fourteen female scientists, followed by a discussion with a journalist and an audience of approximately 300 high school students. ([lien](#))
- **Fondamental ! À la recherche du Futur** 2024
Science outreach video in collaboration with the media outlet L'Esprit Sorcier, founded by Frédéric Courant. Presentation of MRI physics. ([lien](#))
- **Échanges Phocéens Tutor** 2019 - 2021
Tutoring program for middle school students in priority areas of Marseille. Cultural and scientific teaching.